Quadratics & Nonlinear Equations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $\sqrt{2x+1}=5$, what is x?

2 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In the plane, a parabola is given by $f(x)=3(x-4)^2+11$. What is the vertical intercept $f(0)$?

3 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x)=2x^2+x-7$, what is $f(6)$?

4 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x)=2x^2+2x-3$, what is $f(2)$?

5 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x)=x^2-6x-2$, what is $f(5)$?

6 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

An exponential growth model is defined by $f(x)=171\cdot 2^{x/9}$. What is the value $f(18)$?

7 [Foundational] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The product of two consecutive positive integers is 56. What is the smaller integer?

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The graph shows $f(x)=-x^2+4$. What is the maximum value of f?

9 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the larger of the two solutions to $x^2-4x-32=0$?

10 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x)=2x^2+2x-2$, what is $f(5)$?

11 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x)=4x^2+2x+7$, what is $f(-4)$?

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12 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The parabola $y = x^2 - 8x + 15$ intersects the x-axis at $x = 3$ and $x = 5$. What is the x-coordinate of the vertex?

13 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

For what sum of x-values does the graph of $y = x^2 - 3x - 10$ cross the x-axis?

14 [Intermediate] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the sum of the solutions to $x^2 + 5x - 36 = 0$?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the larger of the two solutions to $x^2 + 5x + 6 = 0$?

16 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

Consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. What is the remainder $f(2)$ upon division by $(x - 2)$?

17 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

The quadratic function $f(x) = -x^2 + 12x - 18$ attains its maximum at $x = g$. What is the maximum value $f(g)$?

18 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = x^2 - 5x + 5$, what is $f(-4)$?

19 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $x^2 + 6x + 9 = 25$, what is a possible value of $x + 3$?

20 [Challenging] • [SPR] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

In the partial fraction identity $\frac{6}{(x-11)(x-13)} = \frac{f}{x-13} - \frac{g}{x-11}$, f and g are constants. What is the sum $f + g$?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

Which value of x satisfies $x^2 - 5x + 6 = 0$?

a. $x = 1$

b. $x = 2$

c. $x = 4$

d. $x = 6$

22 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = -2x^2 - 32x - 136$?

a. (-7, -8)

b. (8, -8)

c. (-8, 8)

d. (-8, -8)

23 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = 2x^2 + 2x$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

24 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = 2x^2 + 6x$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

25 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = x^2 - 8x + 21$?

a. (5, 5)

b. (4, 5)

c. (4, -5)

d. (-4, 5)

26 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = x^2 - 2x + 4$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

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27 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = -x^2 + 4x - 4$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

28 [Foundational] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = -x^2 + 5x - 1$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What are the solutions to $x^2 - 5x + 6 = 0$?

a. $x = 1, 6$

b. $x = 2, 3$

c. $x = -2, -3$

d. $x = -1, 6$

30 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = -3x^2 - 42x - 150$?

a. (-7, 3)

b. (-7, -3)

c. (-6, -3)

d. (7, -3)

31 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = -2x^2 - 5x - 3$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

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32 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 4x^2 + 7x$, what is $f(5)$?

a. 135

b. 138

c. 131

d. 139

33 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 5x^2 + 3$, what is $f(4)$?

a. 77

b. 78

c. 76

d. 83

34 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = -3x^2 - 6x + 5$?

a. (-1, 8)

b. (-1, -8)

c. (1, 8)

d. (0, 8)

35 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 4x^2 - x + 9$, what is $f(1)$?

a. 17

b. 20

c. 12

d. 8

36 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 3x^2 - x + 7$, what is $f(-5)$?

a. 87

b. 82

c. 80

d. 94

37 [Intermediate] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the y-intercept of the parabola given by $y = 3x^2 - 7x + 12$?

a. $(0, 3)$

b. $(0, -7)$

c. $(0, 12)$

d. $(12, 0)$

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Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

Which is equivalent to $(x + 4)^2 - 9$?

a. $x^2 + 8x + 7$

b. $x^2 + 8x - 9$

c. $x^2 + 16x + 7$

d. $x^2 + 4x + 7$

39 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = x^2 - 14x + 53$?

a. (7, -4)

b. (-7, 4)

c. (7, 4)

d. (8, 4)

40 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

How many x-intercepts does the graph of $y = -x^2 - x + 2$ have?

a. Zero

b. One

c. Two

d. Cannot be determined

41 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = 2x^2 + 3x + 5$, what is $f(1)$?

a. 10

b. 11

c. 15

d. 4

42 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the larger of the two solutions to $x^2 + x - 72 = 0$?

a. 15

b. 14

c. 8

d. 1

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43 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

What is the vertex of the parabola $y = 2x^2 + 24x + 78$?

a. (-5, 6)

b. (-6, 6)

c. (6, 6)

d. (-6, -6)

44 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = x^2 - 8x + 6$, what is $f(5)$?

a. -9

b. -15

c. -14

d. -18

45 [Challenging] • [MCQ] • Target: Quadratic Equations, Vertex Form, Discriminant & Parabolas

If $f(x) = x^2 - x - 8$, what is $f(6)$?

a. 27

b. 31

c. 22

d. 28

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